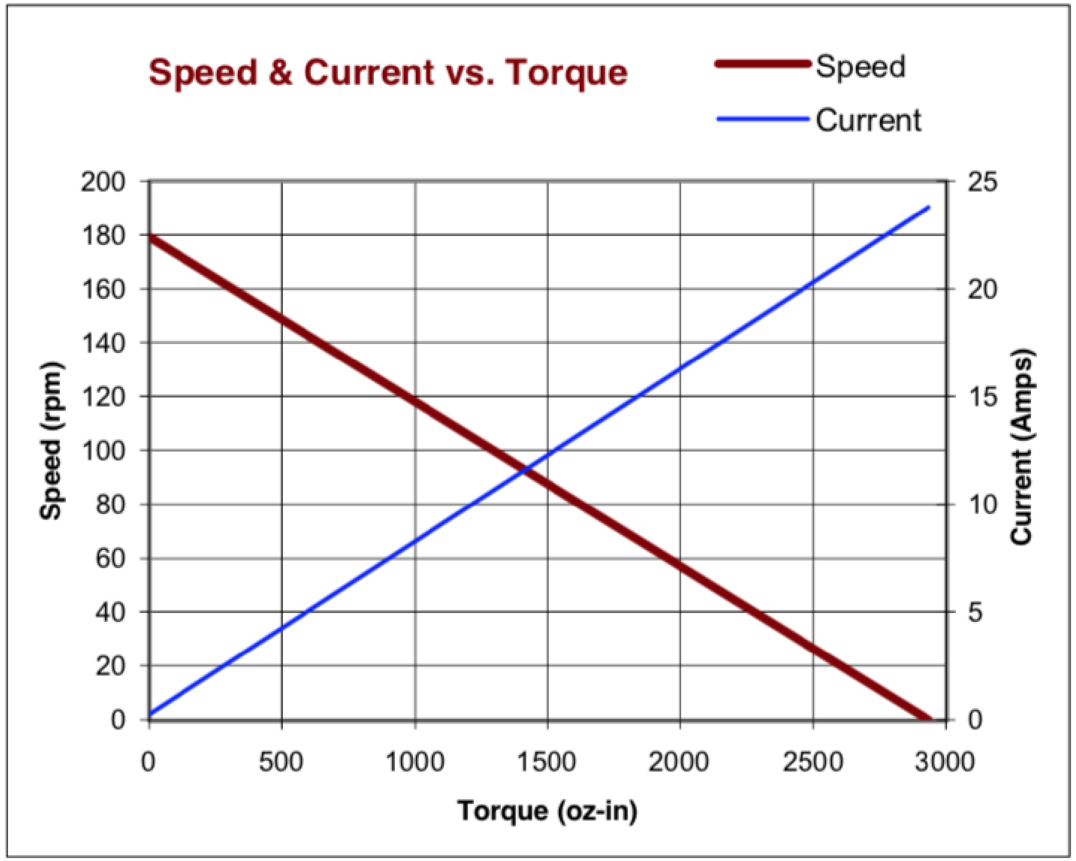
Control 3:
Velocity Control

PID control

$$u(t) = K_P e(t) + K_I \int_0^t e(t)\,dt + K_D \frac{de(t)}{dt}$$

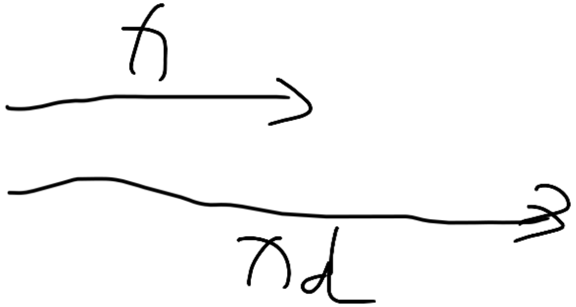
Direct Control of Velocity



$u(t)$ **is PWM**

$$u(t) \approx \frac{dx}{dt} = \dot{x}$$

Remember: Set Point Control



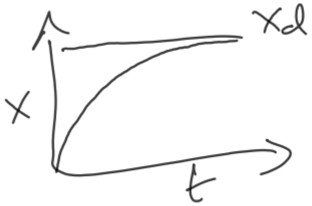
$$e = x_d - x$$

$$\dot{x} = K_p e$$

$$\dot{e} = -\dot{x} = -K_p e$$

$$\dot{e} + K_p e = 0$$

$$e(t) = e(0)\, e^{-K_p t}$$



P Velocity Control

$$\dot{x} = K_p e$$

$$e = x_d - x$$

$$x_d = V_d t, \quad \dot{x}_d = V_d$$ desired speed

$$\dot{e} = \dot{x}_d - \dot{x} = V_d - K_p e$$

or

$$\dot{e} + K_p e = V_d$$

$$e(t) = \frac{V_d}{K_p} + \left(e(0) - \frac{V_d}{K_p}\right) e^{-K_p t}$$

PI Velocity Control

1. PI Control Law

$$\dot{x}(t) = K_p e(t) + K_I \int_0^t e(\tau)\, d\tau$$

2. Error Definition

$$e = x_d - x, \quad \dot{e} = \dot{x}_d - \dot{x}$$

For a constant-velocity trajectory:

$$x_d = V_d t, \quad \dot{x}_d = V_d$$

3. Substitution into Control Law

$$\dot{e} = V_d - (K_p e + K_I \int e\, dt)$$

Differentiating again:

$$\ddot{e} = -K_p \dot{e} - K_I e$$

So the error dynamics become:

$$\ddot{e} + K_p \dot{e} + K_I e = 0$$

4. Characteristic Equation

$$s^2 + K_p s + K_I = 0$$

Solutions:

$$s_{1,2} = -\frac{K_p}{2} \pm \sqrt{\frac{K_p^2}{4} - K_I}$$

- If $\frac{K_p^2}{4} > K_I$: real roots → overdamped, exponential decay.
- If $\frac{K_p^2}{4} = K_I$: critical damping.
- If $\frac{K_p^2}{4} < K_I$: complex roots → underdamped, oscillatory convergence.

Root Locus PI Control

Horizontal axis: **Re(s)** (real part of the poles).

Vertical axis: **Im(s)** (imaginary part of the poles).

Each "×" represents a **pole location** of the closed-loop system as K_I varies.

Case I (far left, real poles):

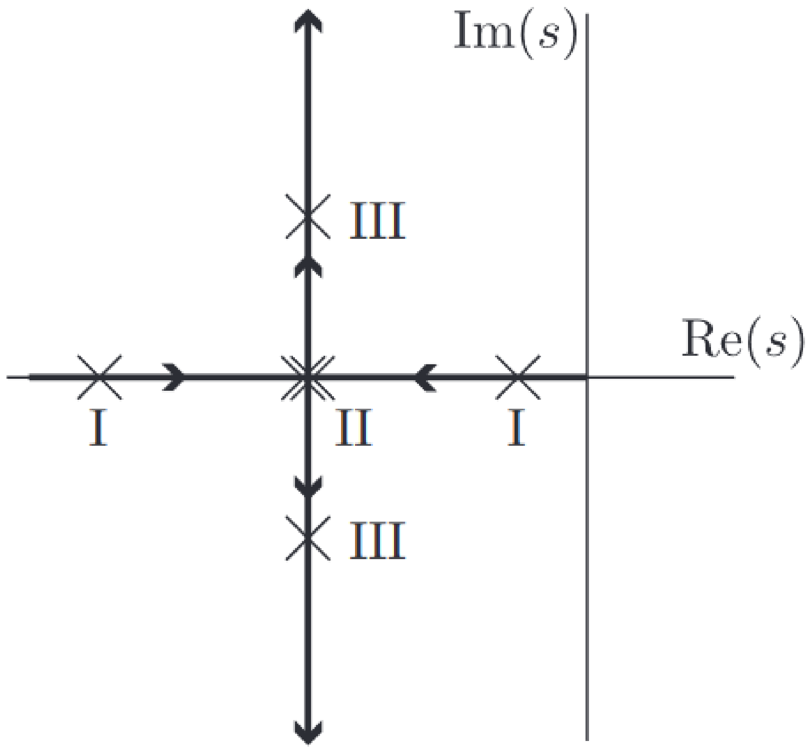
- Overdamped, slow but no oscillation.

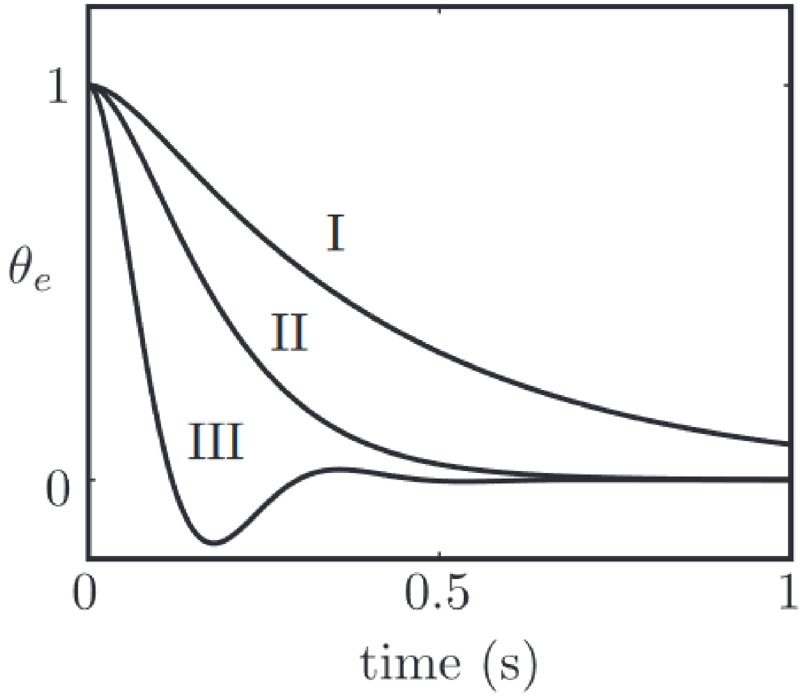
Case II (moderate K_I):

- Faster response, still stable, little overshoot.

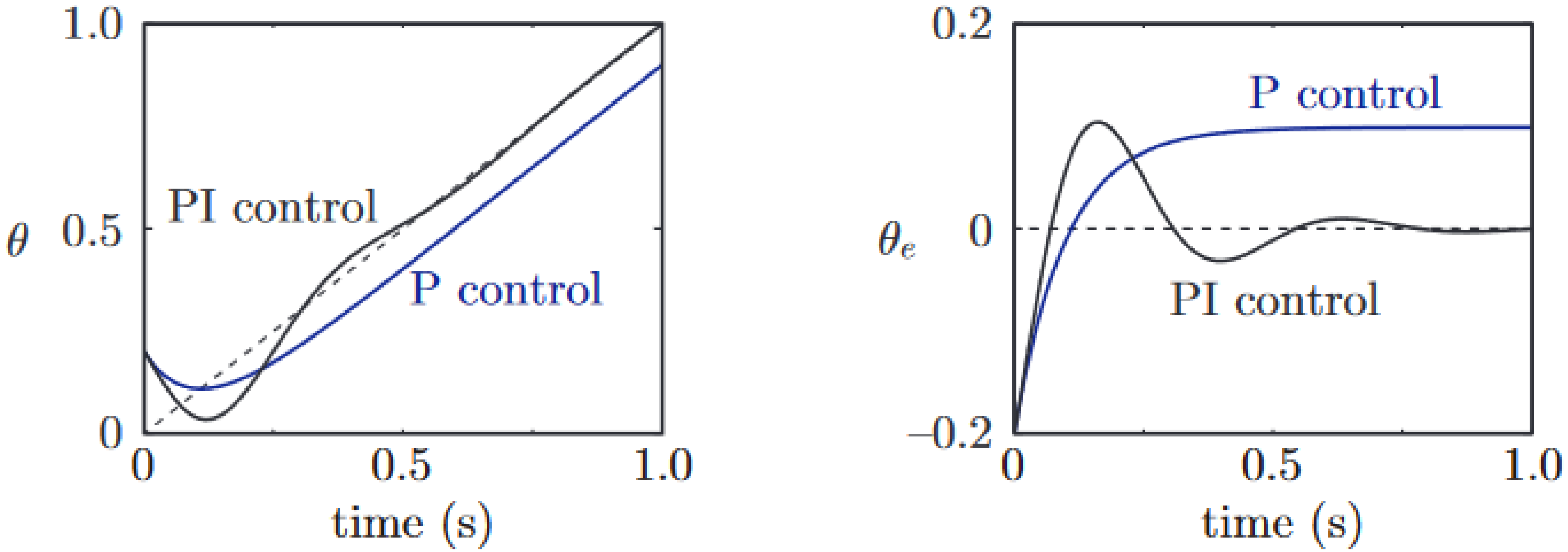
Case III (large K_I):

- Poles complex → oscillations (underdamped).
- Faster initial convergence but with overshoot before settling.





Velocity Control



Aside: Torque control

1. PID Torque Control Law

$$\tau = K_p e + K_I \int e\,dt + K_D \frac{de}{dt}$$

- τ: control input is **joint torque**.

2. Robot Dynamics Equation

$$M\ddot{x} + h(\dot{x}, x) = \tau$$

- M: **mass/inertia matrix** (couples accelerations between joints).
- $h(\dot{x}, x)$: nonlinear effects (Coriolis, centrifugal, gravity).
- $\ddot{x}$: acceleration.
- τ: applied joint torques.

When the PID law is applied:

$$M\ddot{x} + h(\dot{x}, x) = K_p e + K_I \int e\,dt + K_D \frac{de}{dt}$$

3. Expanded Example (2-DOF Robot Arm)

The lower part shows the **mass matrix expansion** for a 2-joint system:

$$\begin{bmatrix} m_1 & m_2 \\ m_2 & m_1 + m_2 \end{bmatrix} \begin{bmatrix} \ddot{\theta}_1 \\ \ddot{\theta}_2 \end{bmatrix} + h(\dot{\theta}, \theta) = \begin{bmatrix} \tau_1 \\ \tau_2 \end{bmatrix}$$

- This illustrates how accelerations of each joint depend on both joints (dynamic **coupling**).

Correct answer

Question 2

10 / 10 pts

What is the base 10 value of the **signed** (represented using two's complement) binary integer 0b1001001110100101?

-27,739

-27,739 (with margin: 0)

Additional Comments:

Question 11

5 / 10 pts

The flash memory on the ATmega328P is used for

- [x] user programs
- [ ] data
- [x] Boot program
- [x] Interrupt vectors

Additional Comments:

Trapezoidal Velocity Profile

- Next time

Differential Drive Robot Control